

Algebra II Final Topics Review

Instructions

Complete in 90 minutes. Show clear work. Use exact values unless a decimal approximation is explicitly requested.

Reference Facts

- Log properties:

$$\log_b(MN) = \log_b M + \log_b N, \quad \log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N, \quad \log_b(M^r) = r \log_b M$$

- Hyperbola forms:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \quad \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

- Arithmetic and geometric sequences:

$$a_n = a_1 + (n-1)d, \quad a_n = a_1 r^{n-1}$$
$$S_n = \frac{n(a_1 + a_n)}{2}, \quad S_n = \frac{a_1(1-r^n)}{1-r}, \quad S_\infty = \frac{a_1}{1-r} \quad (|r| < 1)$$

- Counting:

$$n! = n(n-1)(n-2) \cdots 1, \quad {}_n P_r = \frac{n!}{(n-r)!}, \quad {}_n C_r = \frac{n!}{r!(n-r)!}$$

- Probability:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B), \quad P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$$

- Statistics:

$$z = \frac{x - \mu}{\sigma}, \quad \text{range} = \max - \min, \quad \text{IQR} = Q_3 - Q_1$$

Problems

1. Condense into a single logarithm:

$$3 \log_2(x) - \frac{1}{2} \log_2(y+1) + 2 \log_2(4) - \log_2(z^3) + \log_2\left(\frac{w}{2}\right).$$

2. Expand completely:

$$\log_b\left(\frac{5x^3\sqrt{y}}{b^2\sqrt[3]{z^4}}\right).$$

3. Solve and check for extraneous solutions:

$$\log_3(x-1) + \log_3(x-7) = 2 + \log_3(5).$$

4. Solve and state the domain restrictions used:

$$\log_5(3x+2) - \log_5(x-4) = 1.$$

5. Given

$$\log_2(a) = m, \quad \log_2(b) = n,$$

write $\log_2\left(\frac{8a^3}{\sqrt{b}}\right)$ in terms of m and n .

6. Determine whether each statement is always true or sometimes false. If false, give a correction or counterexample.

$$\log_b(x+y) = \log_b x + \log_b y, \quad \log_b(x^4) = 4 \log_b x, \quad b^{\log_b x} = x.$$

7. Find the inverse function of

$$f(x) = 2 \log_3(x-4) - 5.$$

State the domain of f and the domain of f^{-1} .

8. Graph the key features of

$$y = \log_2(x+3) - 4.$$

State the vertical asymptote, domain, range, and x -intercept.

9. A bank account earns interest compounded annually and is modeled by

$$A(t) = 1800(1.045)^t.$$

How long will it take the account to double? Then find how long it takes to reach \$3,000. Round each to the nearest tenth of a year.

10. Put the hyperbola in standard form and identify its center, orientation, vertices, and asymptotes:

$$9x^2 - 16y^2 - 54x - 64y - 127 = 0.$$

11. Write the equation of the hyperbola with center $(2, -1)$, vertices $(2, 4)$ and $(2, -6)$, and asymptotes

$$y + 1 = \pm \frac{5}{3}(x - 2).$$

Then find the coordinates of the foci.

12. For the arithmetic sequence with $a_5 = 18$ and $S_{21} = 777$, find a_1 , d , and a_{50} .

13. Find the exact value:

$$\sum_{k=1}^{40} (7 - 3k).$$

14. For a geometric sequence, $a_3 = 24$ and $a_7 = \frac{3}{2}$. Find all possible values of r and a_1 , then find a_{10} for each possible sequence.

15. Evaluate exactly:

$$\sum_{k=0}^8 6 \left(\frac{2}{3} \right)^k.$$

16. Determine whether the infinite series converges. If it converges, find its exact sum:

$$\sum_{n=1}^{\infty} 12 \left(-\frac{1}{4} \right)^{n-1}.$$

17. Convert the repeating decimal to a fraction:

$$0.1\overline{27}.$$

18. Write a recursive formula and an explicit formula for the sequence:

$$7, 3, -1, -5, \dots$$

19. Rewrite using sigma notation:

$$4 + 8 + 16 + 32 + \dots + 512.$$

20. How many different arrangements can be made from the letters in the word LEVEL if the two E's cannot be adjacent?

21. A club has 9 juniors and 7 seniors. How many committees of 5 students can be formed if the committee must include at least 2 juniors and at least 2 seniors?
22. A password must contain 2 different letters followed by 3 digits. Letters may not repeat, digits may not repeat, and the first digit cannot be 0. How many passwords are possible?
23. A standard deck of 52 cards is shuffled and one card is drawn. Find:

$$P(\text{heart or king}), \quad P(\text{not a face card}).$$

24. Two fair six-sided dice are rolled. Find the probability that the sum is at least 9 or the two dice show the same number.
25. A bag contains 5 red, 4 blue, and 3 green marbles. Three marbles are drawn without replacement. Find the probability that all three are the same color.
26. The table shows survey results from 100 students.

	Likes Math	Does Not Like Math	Total
Takes Art	18	22	40
Does Not Take Art	27	33	60
Total	45	55	100

Find $P(\text{likes math} \mid \text{takes art})$ and $P(\text{takes art} \mid \text{likes math})$. Then determine whether “likes math” and “takes art” are independent.

27. A data set has mean 72 and standard deviation 8. Find the z -score for $x = 86$. Then find the data value with $z = -1.25$.
28. The data set is
- $$12, 15, 16, 18, 20, 24, 28, 30, 35.$$

Find the median, Q_1 , Q_3 , range, and IQR.

29. A normally distributed test has mean 80 and standard deviation 6. Use the empirical rule to estimate the percent of scores between 68 and 92, then estimate the percent above 92.
30. Classify each as observational study or experiment:
- A researcher records how many hours students sleep and compares that to quiz scores.
 - A researcher randomly assigns students to two study methods and compares quiz scores.

31. Find the amplitude, period, midline, and phase shift of

$$y = -3 \sin \left(2x - \frac{\pi}{2} \right) + 1.$$

Then state the range.

32. Solve on $0 \leq x < 2\pi$:

$$2 \cos^2 x - \cos x - 1 = 0.$$